Particle Physics Note

Author: Lingyu Xia

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Chapter 1

Decay Rate & Cross Sections

§ 1.1 Fermi Golden Rule

Consider time-dependent Schrödinger equation,

$$i\frac{d\psi}{dt} = \hat{H}(t)\psi. \tag{1.1}$$

We can separate Hamiltonian operator into two parts, one is time-independent, and else is time-dependent,

$$\hat{H}(t) = \hat{H}_0 + \hat{H}'(\mathrm{x}, t) \tag{1.2}$$

For time-independent part we can use common method to solve it,

$$\hat{H}_0\psi_k = E_k\psi_k, \quad \text{and} \quad \langle\psi_j|\psi_k\rangle = \delta_{jk} \tag{1.3}$$

Then the wavefunction can be expressed in terms of complete set of states of the unperturbed Hamiltonian as

$$\psi(\mathrm{x}, t) = \sum_k c_k(t)\psi(\mathrm{x})e^{-iE_k t}. \tag{1.4}$$

Then we can substitute Eq. 1.4 into Eq. 1.1, we can get

$$\begin{aligned} i\sum_k \left[\frac{\mathrm{d}c_k}{\mathrm{d}t}\psi_k e^{-iE_k t} - iE_k c_k \psi_k e^{-iE_k t}\right] &= \sum_k c_k \hat{H}_0 \psi_k e^{-iE_k t} + \sum_k \hat{H}' c_k \psi_k e^{-iE_k t} \\ \Rightarrow \quad i\sum_k \frac{\mathrm{d}c_k}{\mathrm{d}t}\psi_k e^{-iE_k t} &= \sum_k \hat{H}' c_k(t)\psi_k e^{-iE_k t} \end{aligned} \tag{1.5}$$

Suppose the initial state is i for $\psi_i \equiv |i\rangle$, that is

$$\psi(\mathrm{x}, 0) = \sum_k c_k(0)\psi_k = c_i(0)\psi_i. \tag{1.6}$$

We can easily imply that $c_k = \delta_{ik}$. Assume that time-dependent Hamiltonian is perturbed, which implies $c_{i\neq k} \ll 1, c_i \sim 1$. And Eq. 1.5 can reduce to

$$i \sum_k \frac{\mathrm{d}c_k}{\mathrm{d}t} |k\rangle e^{-iE_k t} \approx \hat{H}' |i\rangle e^{-iE_i t} \tag{1.7}$$

Then we need to get to final state, so we can use relation of inner product, let $\langle f|$ act on Eq. 1.7, we can get the coefficient c_f

$$\frac{dc_f}{dt} = -i \langle f|\hat{H}'|i\rangle e^{-i(E_i - E_f)t}, \tag{1.8}$$

where $\langle f|\hat{H}'|i\rangle \equiv T_{fi}$ is **transition matrix element**, can be calculated by

$$\langle f|\hat{H}'|i\rangle = \int_V \psi_f^\dagger(\mathrm{x}) \hat{H}' \psi_i(\mathrm{x}) \, \mathrm{d}^3\mathrm{x}. \tag{1.9}$$

The probability for a transition to the state $|f\rangle$ is given by

$$P_{if} = c_f^\dagger(t) c_f(t) \tag{1.10}$$

§ 1.2
Decay Rate

§ 1.3
Cross Sections

Chapter 2

Klein-Gordon Equation